

Report for setting up a camera in raspberry pi 4

CE6461 – Computer Vision Systems

Student Name : **L.Sharon Danish Laus**

Student ID : **25249347**

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Introduction:

This report outlines the setup and verification of a computer vision platform consisting of a Raspberry Pi 4 and outlines the steps to move it to a networked remote access over Ethernet. The work preconditions the further laboratories with vision because it prepares the RPi4 kit that includes a built-in camera and proves the build with the help of the photo documentation that includes the Student ID card, the setting up of the network parameters, and the implementation of the command-line access via SSH.

The kit is first assembled as per the instructions given, and all the procedures involved in connecting the camera module and related peripherals are handled carefully. The next steps are configuring and test connectivity to the network by means of obtaining the Raspberry Pi Ethernet and Wi-Fi MAC addresses via the console that identifies the device in the laboratory network. Finally, the direct wired connection is found to be effective because it is possible to open an SSH session to the Raspberry Pi via Ethernet, and a terminal screenshot is registered as the empirical evidence.

These steps altogether ensure that the hardware is properly assembled, and the device is uniquely identifiable on the network, and the secure remote access is provided. The resulting artefacts, i.e. a

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photograph of the put-together kit with the Student ID card, a console screenshot of the MAC addresses demarcation, and a terminal screenshot of a SSH login are verifiable checkpoints that show that the platform is ready to be experimented with computer vision.

Network Configuration:

```
File Edit Tabs Help
s25249347@raspberrypi:~ $ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc mq state DOWN group default qlen 1000
    link/ether d8:3a:dd:af:6a:18 brd ff:ff:ff:ff:ff:ff
3: wlan0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether d8:3a:dd:af:6a:19 brd ff:ff:ff:ff:ff:ff
    inet 10.153.133.33/24 brd 10.153.133.255 scope global dynamic noprefixroute wlan0
        valid_lft 3349sec preferred_lft 3349sec
    inet6 fe80::a3bf:1ec1:6652:69f/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
s25249347@raspberrypi:~ $
```

Camera Setup and Media Capture:

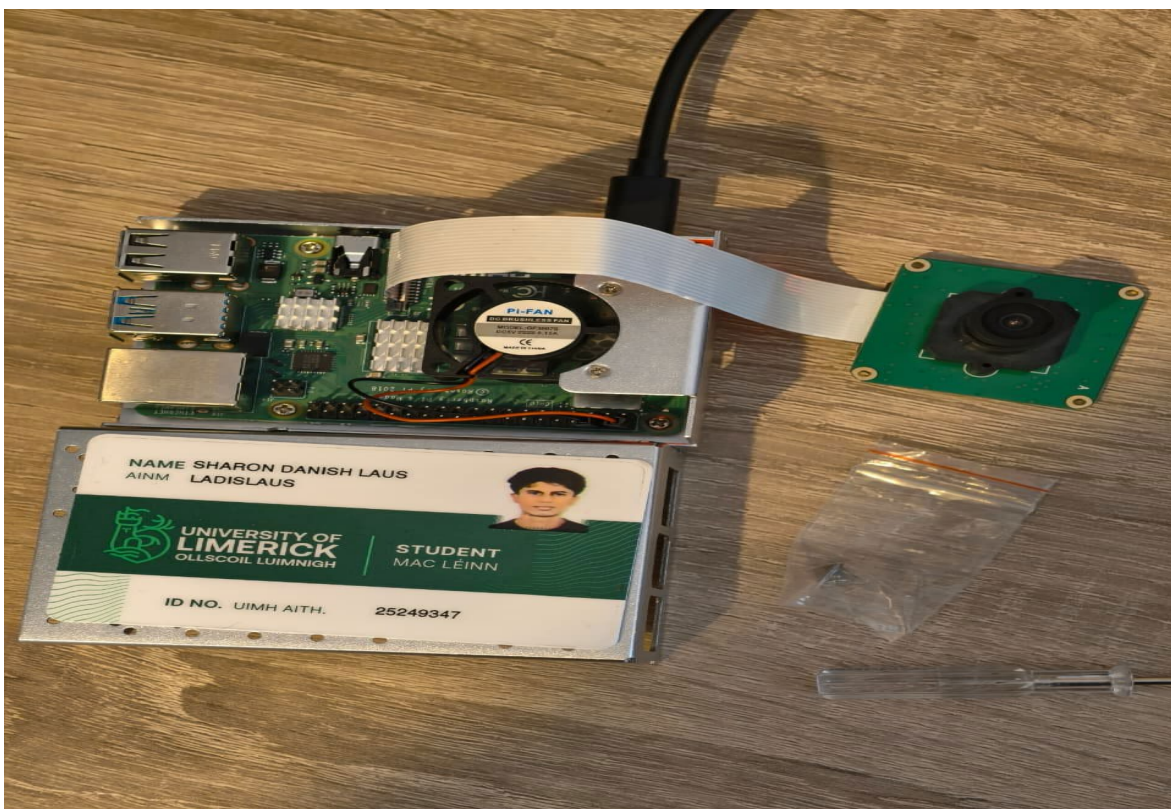


Figure 1:
Assembled
Raspberry Pi
4 kit

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Figure 2: Still image captured on the Raspberry Pi 4 camera

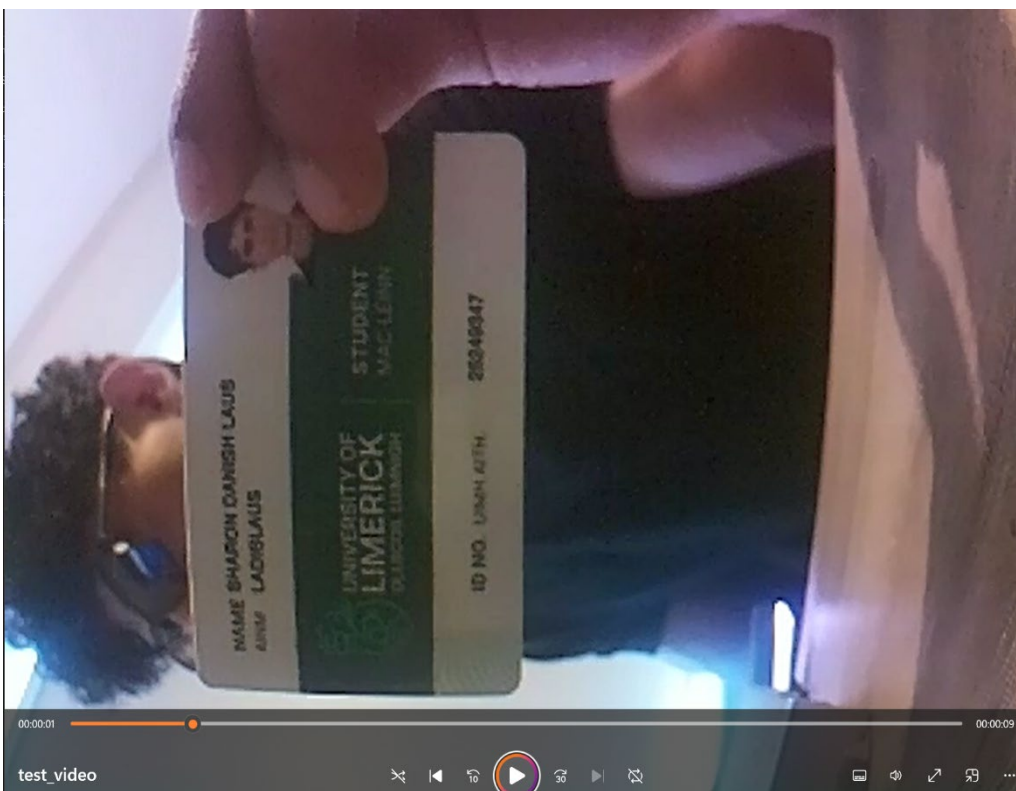


Figure 3: Screenshot of a video frame

Image Quality Analysis:

Exposure and clarity: All sharpness is moderate and not crisp, sides of the card are slightly soft, The Highlights on the bottom a slight clipping of the visible lamp or reflection, and midtones on the card are readable. Luminance noise is low in flat regions indicating that the scene was well illuminated or the pipeline de-noised it.

Colour and tone: Nominally neutral areas have a little tint of magenta, suggesting that the mixed indoor lighting influenced auto white balance. There is sufficient contrast but it may be enhanced with increased distance between the bright source of light and the subject to eliminate veiling glare.

Improvements:

- 1)Adjust the indoor white-balance mode or even manual red/blue gains to the neutral in the case of tungsten or fluorescent light to remove the magenta cast.
- 2)Turn down the analog sensitivity (which is analogous to the ISO) and allow the exposure period to creep up, or simply set a shutter speed of about 1/60-1/125 s to use with the handheld.
- 3)Move the subject slightly back to put him in the sharper range of the fixed-focus module, which gives better acuity to the card surface.
- 4)Avoid bright light sources in the frame as it may highlight clipping and glare onto the card.
- 5)Record the sensor at full still resolution (2592x1944) to record more detail and be able to do more with cropping and rotatio

Camera Comparison:

Identification and Specifications of camera modules:

The OKDO 5MP Camera uses the OmniVision OV5647 sensor, which is a rolling shutter CMOS image sensor used in consumer electronics. The sensor has the 1/4-inch optical format with 2592x1944 pixel count (5 megapixels overall), which produces 1.4 mm pixels. Video capturing is supported at 1080p30 and 720p60 frame rates with fixed 90 deg diagonal field of view of f/2.2 aperture. It uses the OmniBSI (backside illumination) technology to enhance light sensitivity and sends raw RGB data using MIPI CSI-2 or parallel digital video interface.

Comparison to Automotive Camera Systems:

Advanced driver Assistance Systems (ADAS) and autonomous driving auto-motive cameras make use of much more sophisticated sensor technology. Sony IMX390 and IMX490 sensors are standardized in the industry, with a format size of 1/1.55 inch, 5.4 megapixels and 3.0 mm pixel sizes - over two times the size of the pixels in the OV5647. These car sensors reach 120dB high dynamic range (HDR) with digital overlap algorithms, in contrast to the around 60dB native range of the OV5647. Other auto-specific design features of interest are LED flicker rejection to detect traffic lights, global or pseudo-global shutter operation to avoid rolling shutter artifacts, and operation in the temperature range of -40 C to +105 C. The cameras are subjected to extreme automotive testing such as ISO 16750 vibration/shock resistance, and IP67/IP69K ingress protection. These are typically interfaced through GMSL2 (Gigabit Multimedia Serial Link) on longer cable runs and electromagnetic compatibility, and typically with multi-camera arrays using special automotive processors such as NVIDIA Jetson platforms.

Comparison to Automation System in Manufacturing Cameras:

In robot vision camera, industrial world focuses on global shutter to avoid motion artifacts in image capturing of objects that move at a high pace on assembly lines. The Sony IMX250 (5.01 MP) and OmniVision OG09A (9 MP) have 3.45 mm pixels, with global shutter architecture and 60-155 fps full-resolution functionality. These cameras have ruggedized housings that are rated IP67 or above in terms of dust and moisture protection in rough factory environments. Industrial cameras use CoaXPress or GigE Vision interfaces that offer deterministic data transfer to be used in real-time machine vision processing. With external hardware triggers, frame synchronization provides multi-camera stereo vision systems to robotic pick-and-place tasks. The sensors generally provide 60-76dB dynamic range enough to work in controlled lighting scenarios but provide other features such as dual conversion gain to enhance low light operation.

Similarities, Differences and Major Limitations:

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Those are similarities such as CMOS sensor technology, support of MIPI CSI-2 output, and resolution in megapixels that is compatible with fundamental imaging functions. The three types of cameras use backside illumination to enhance quantum efficiency.

Distinctions are significant: automotive and industrial cameras have 2-3x larger pixels to allow higher light sensitivity, can operate in global shutter mode to ensure no distortion is created by motion, have a broader temperature range (-40 deg C to +85 deg C industrial or -40 deg C to +105 deg C automotive versus 0 deg C to +60 deg C consumer), and have dedicated interfaces to ensure they are electromagnetically immune. Professional systems can achieve 2-4x the frame rate at the same resolutions and use hardware-based HDR in the 100dB range over the software-limited range of the OV5647.

Central Constraints of the OKDO 5MP camera to professional use encompass: rolling shutter artifacts making it unsuitable to use in high-speed vehicle or robotic applications where distortion can cause perception errors, inadequate dynamic range to use in automotive applications with high contrast between shadows and direct sunlight, (c) lack of environmental robustness, e.g. against temperature extremes, vibration, and exposure to moisture, (d) limited frame rate to real-time processing of high-speed assembly lines, (e) small pixel size negatively affecting low-light sensitivity needed in nighttime

Conclusion:

This lab experiment was able to demonstrate the physical assembly and setup of an OKDO 5 mega pixel camera with an OV5647 sensor on the Raspberry Pi platform including the establishment of wireless connectivity, connecting a CSI cable, and running command line image/video capture through libcamera tools. The practical experience of the hands-on installation process with camera module integration and the standard file transfer protocols needed in computer-vision applications was quite useful. Even though the OKDO 5-MP camera with OV5647 sensor provides enough resolution and performance to meet the instructional computer-vision needs, a comparative analysis found some basic limitations that excludes its use in professional industrial applications. These are severe limitations such as a lack of dynamic range (68 dB as compared to the greater than 100 dB needed in automotive applications), rolling-shutter effects that are unacceptable in an autonomous vehicle, consumer-grade temperature limits, and lack of specialist colour-processing features needed by medical diagnostics. In its turn, this assignment demonstrates that although the OV5647 sensor is an effective educational tool to understand the fundamentals of the camera, the fundamentals of image acquisition, and the fundamentals of integrating an embedded-system, professional use in both automotive and medical imaging requires purpose-specific sensors with improved performance characteristics, regulatory compliance, and hazardous design, that significantly exceed the capabilities of consumer products.

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